10

Sparse Recovery

In this chapter we focus entirely on applications of high-dimensional probability to data science. We study basic signal recovery problems in compressed sensing and structured regression problems in high-dimensional statistics, and we develop algorithmic methods to solve them using convex optimization.

We introduce these problems in Section 10.1. Our first approach to them, which is very simple and general, is developed in Section 10.2 based on the M^* bound. We then specialize this approach to two important problems. In Section 10.3 we study the sparse recovery problem, in which the unknown signal is sparse (i.e. has few non-zero coordinates). In Section 10.4, we study low-rank matrix recovery problem, in which the unknown signal is a low-rank matrix. If instead of M^* bound we use the escape theorem, it is possible to do *exact* recovery of sparse signals (without any error)! We prove this basic result in compressed sensing in Section 10.5. We first deduce it from the escape theorem, and then we study an important deterministic condition that guarantees sparse recovery – the restricted isometry property. Finally, in Section 10.6 we use matrix deviation inequality to analyze Lasso, the most popular optimization method for sparse regression in statistics.

10.1 High-dimensional signal recovery problems

Mathematically, we model a *signal* as a vector $x \in \mathbb{R}^n$. Suppose we do not know x, but we have m random, linear, possibly noisy *measurements* of x. Such measurements can be represented as a vector $y \in \mathbb{R}^m$ with following form:

$$y = Ax + w. \tag{10.1}$$

Here A is an $m \times n$ known measurement matrix and $w \in \mathbb{R}^m$ is an unknown *noise* vector; see Figure 10.1. Our goal is to recover x from A and y as accurately as possible.

Note that the measurements $y = (y_1, \ldots, y_m)$ can be equivalently represented as

$$y_i = \langle A_i, x \rangle + w_i, \quad i = 1, \ldots, m \tag{10.2}$$

where $A_i \in \mathbb{R}^n$ denote the rows of the matrix A.

To make the signal recovery problem amenable to the methods of high-dimensional probability, we assume the following probabilistic model. We suppose that the measurement matrix A in (10.1) is a realization of a random matrix drawn from some distribution. More specifically, we assume that the rows A_i are independent random vectors, which makes the observations y_i independent, too.

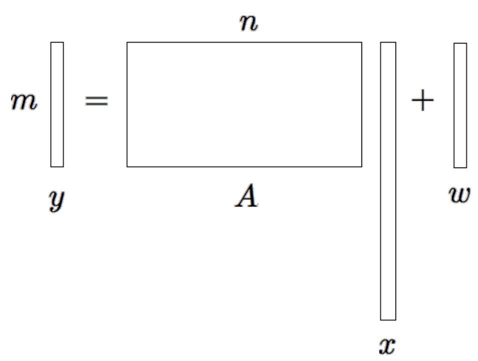


Figure 10.1 Signal recovery problem: recover a signal x from random, linear measurements y.

Example 10.1.1 (Audio sampling)**.** In signal processing applications, x can be a digitized audio signal. The measurement vector y can be obtained by sampling x at m randomly chosen time points, see Figure 10.2.

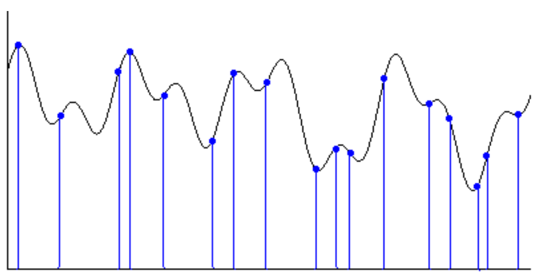


Figure 10.2 Signal recovery problem in audio sampling: recover an audio signal x from a sample of x taken at m random time points.

Example 10.1.2 (Linear regression)**.** The linear regression is one of the major inference problems in Statistics. Here we would like to model the relationship between n *predictor variables* and a *response variable* using a sample of m observations. The regression problem is usually written as

$$Y = X\beta + w.$$

Here X is an $m \times n$ matrix that contains a sample of predictor variables, $Y \in \mathbb{R}^m$ is a vector that contains a sample of response variables, $\beta \in \mathbb{R}^n$ is a coefficient vector that specifies the relationship that we try to recover, and w is a noise vector.

For example, in genetics one could be interested in predicting a certain disease based on genetic information. One then performs a study on m patients collecting the expressions of their n genes. The matrix X is defined by letting X_{ij} be the expression of gene j in patient i, and the coefficients Y_i of the vector Y can be set to quantify whether or not patient i has the disease (and to what extent). The goal is to recover the coefficients of β, which quantify how each gene affects the disease.

10.1.1 Incorporating prior information about the signal

Many modern signal recovery problems operate in the regime where

$$m \ll n,$$

i.e. we have far fewer measurements than unknowns. For instance, in a typical genetic study like the one described in Example 10.1.2, the number of patients is ~ 100 while the number of genes is $\sim 10,000$.

In this regime, the recovery problem (10.1) is *ill-posed* even in the noiseless case where $w = 0$. It can not be even approximately solved: the solutions form a linear subspace of dimension at least $n - m$. To overcome this difficulty, we can leverage some *prior information* about the signal x – something that we know, believe, or want to enforce about x. Such information can be mathematically be expressed by assuming that

$$x \in T \tag{10.3}$$

where $T \subset \mathbb{R}^n$ is a known set.

The smaller the set T, the fewer measurements m could be needed to recover x. For small T, we can hope that signal recovery can be solved even in the ill-posed regime where $m \ll n$. We see how this idea works in the next sections.

10.2 Signal recovery based on M^* bound

Let us return to the the recovery problem (10.1). For simplicity, let us first consider the noiseless version of the problem, that is

$$y = Ax, \quad x \in T.$$

To recap, here $x \in \mathbb{R}^n$ is the unknown signal, $T \subset \mathbb{R}^n$ is a known set that encodes our prior information about x, and A is a known $m \times n$ random measurement matrix. Our goal is to recover x from y.

Perhaps the simplest candidate for the solution would be *any* vector x' that is consistent both with the measurements and the prior, so we

$$\text{find } x' : \; y = Ax', \quad x' \in T. \tag{10.4}$$

If the set T is convex, this is a convex program (in the feasibility form), and many effective algorithms exists to numerically solve it.

This naïve approach actually works well. We now quickly deduce this from the M^* bound from Section 9.4.1.

Theorem 10.2.1. *Suppose the rows A_i of A are independent, isotropic and sub-gaussian random vectors. Then any solution $\widehat{x}$ of the program (10.4) satisfies*

$$\mathbb{E}\,\|\widehat{x}-x\|_2 \le \frac{CK^2 w(T)}{\sqrt{m}},$$

where $K = \max_i \|A_i\|_{\psi_2}$.

Proof Since $x, \widehat{x} \in T$ and $Ax = A\widehat{x} = y$, we have

$$x, \widehat{x} \in T \cap E_x$$

where $E_x := x + \ker A$. (Figure 10.3 illustrates this situation visually.) Then the

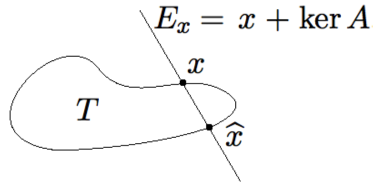


Figure 10.3 Signal recovery: the signal x and the solution $\widehat{x}$ lie in the prior set T and in the affine subspace E_x.

affine version of the M^* bound (Exercise 9.4.3) yields

$$\mathbb{E}\,\|\widehat{x}-x\|_2 \le \mathbb{E}\,\mathrm{diam}(T\cap E_x) \le \frac{CK^2 w(T)}{\sqrt{m}}.$$

This completes the proof. □

Remark 10.2.2 (Stable dimension)**.** Arguing as in Remark 9.4.4, we obtain a non-trivial error bound

$$\mathbb{E}\,\|\widehat{x}-x\|_2 \le \varepsilon\cdot\mathrm{diam}(T)$$

provided that the number of measurements m is so that

$$m \ge C(K^4/\varepsilon^2)d(T).$$

In words, *the signal can be approximately recovered as long as the number of measurements m exceeds a multiple of the stable dimension $d(T)$ of the prior set T.*

Since the stable dimension can be much smaller than the ambient dimension n, the recovery problem may often be solved even in the high-dimensional, ill-posed regime where

$$m \ll n.$$

We see some concrete examples of this situation shortly.

Remark 10.2.3 (Convexity)**.** If the prior set T is not convex, we can convexify it by replacing T with its convex hull $\mathrm{conv}(T)$. This makes (10.4) a convex program, and thus computationally tractable. At the same time, the recovery guarantees of Theorem 10.2.1 do not change since

$$w(\mathrm{conv}(T)) = w(T)$$

by Proposition 7.5.2.

Exercise 10.2.4 (Noisy measurements)**.** ☕☕ Extend the recovery result (Theorem 10.2.1) for the noisy model $y = Ax + w$ we considered in (10.1). Namely, show that

$$\mathbb{E}\,\|\widehat{x} - x\|_2 \le \frac{CK^2 w(T) + \|w\|_2}{\sqrt{m}}.$$

Hint: Modify the argument that leads to the M^* bound.

Exercise 10.2.5 (Mean squared error)**.** ☕☕☕ Prove that the error bound Theorem 10.2.1 can be extended for the mean squared error

$$\mathbb{E}\,\|\widehat{x} - x\|_2^2.$$

Hint: Modify the M^* bound accordingly.

Exercise 10.2.6 (Recovery by optimization)**.** Suppose T is the unit ball of some norm $\|\cdot\|_T$ in $\mathbb{R}^n$. Show that the conclusion of Theorem 10.2.1 holds also for the solution of the following optimization program:

$$\text{minimize } \|x'\|_T \text{ s.t. } y = Ax'.$$

10.3 Recovery of sparse signals

10.3.1 Sparsity

Let us give a concrete example of a prior set T. Very often, we believe that x should be *sparse*, i.e. that most coefficients of x are zero, exactly or approximately. For instance, in genetic studies like the one we described in Example 10.1.2, it is natural to expect that very few genes (~ 10) have significant impact on a given disease, and we would like to find out which ones.

In some applications, one needs to change the basis so that the signals of interest are sparse. For instance, in the audio recovery problem considered in Example 10.1.1, we typically deal with *band-limited* signals x. Those are the signals whose frequencies (the values of the Fourier transform) are constrained to some small set, such as a bounded interval. While the audio signal x itself is not sparse as is apparent from Figure 10.2, the Fourier transform of x may be sparse. In other words, x may be sparse in the frequency and not time domain.

To quantify the (exact) sparsity of a vector $x \in \mathbb{R}^n$, we consider the size of the support of x which we denote

$$\|x\|_0 := |\,\mathrm{supp}(x)| = \big|\{i:\ x_i \neq 0\}\big|\,.$$

Assume that

$$\|x\|_0 = s \ll n. \tag{10.5}$$

This can be viewed as a special case of a general assumption (10.3) by putting

$$T = \left\{x \in \mathbb{R}^n : \ \|x\|_0 \le s\right\}.$$

Then a simple dimension count shows the recovery problem (10.1) could become well posed:

Exercise 10.3.1 (Sparse recovery problem is well posed)**.** ☕☕☕ Argue that if A is in general position and $m \ge 2\|x\|_0$, the solution to the sparse recovery problem (10.1) is unique if it exists. (Choose a useful definition of general position for this problem.)

Even when the problem (10.1) is well posed, it could be computationally hard. It is easy if one knows the support of x (why?) but usually the support is unknown. An exhaustive search over all possible supports (subsets of a given size s) is impossible since the number of possibilities is exponentially large: $\binom{n}{s} \ge 2^s$.

Fortunately, there exist computationally effective approaches to high-dimensional recovery problems with general constraints (10.3), and the sparse recovery problems in particular. We cover these approaches next.

Exercise 10.3.2 (The "ℓ_p norms" for $0 \le p < 1$)**.** ☕☕☕

(a) Check that $\|\cdot\|_0$ is not a norm on $\mathbb{R}^n$.
(b) Check that $\|\cdot\|_p$ is not a norm on $\mathbb{R}^n$ if $0 < p < 1$. Figure 10.4 illustrates the unit balls for various ℓ_p "norms".
(c) Show that, for every $x \in \mathbb{R}^n$,

$$\|x\|_0 = \lim_{p \to 0_+} \|x\|_p^p.$$

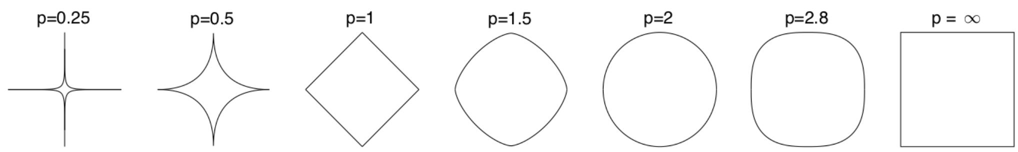


Figure 10.4 The unit balls of ℓ_p for various p in $\mathbb{R}^2$.

10.3.2 Convexifying the sparsity by ℓ_1 norm, and recovery guarantees

Let us specialize the general recovery guarantees developed in Section 10.2 to the sparse recovery problem. To do this, we should choose the prior set T so that it promotes sparsity. In the previous section, we saw that the choice

$$T := \left\{x \in \mathbb{R}^n : \ \|x\|_0 \le s\right\}$$

does not allow for computationally tractable algorithms.

To make T convex, we may replace the "ℓ_0 norm" by the ℓ_p norm with the smallest exponent $p > 0$ that makes this a true norm. This exponent is obviously $p = 1$ as we can see from Figure 10.4. So let us repeat this important heuristic: *we propose to replace the ℓ_0 "norm" by the ℓ_1 norm.*

Thus it makes sense to choose T to be a scaled ℓ_1 ball:

$$T := \sqrt{s} B_1^n.$$

The scaling factor $\sqrt{s}$ was chosen so that T can accommodate all s-sparse unit vectors:

Exercise 10.3.3. ☕ Check that

$$\left\{x \in \mathbb{R}^n : \ \|x\|_0 \le s, \ \|x\|_2 \le 1\right\} \subset \sqrt{s} B_1^n.$$

For this T, the general recovery program (10.4) becomes

$$\text{Find } x' : \ y = Ax', \quad \|x'\|_1 \le \sqrt{s}. \tag{10.6}$$

Note that this is a convex program, and therefore is computationally tractable. And the general recovery guarantee, Theorem 10.2.1, specialized to our case, implies the following.

Corollary 10.3.4 (Sparse recovery: guarantees)**.** *Assume the unknown s-sparse signal $x \in \mathbb{R}^n$ satisfies $\|x\|_2 \le 1$. Then x can be approximately recovered from the random measurement vector $y = Ax$ by a solution $\widehat{x}$ of the program* (10.6)*. The recovery error satisfies*

$$\mathbb{E}\, \|\widehat{x} - x\|_2 \le CK^2 \sqrt{\frac{s \log n}{m}}.$$

Proof Set $T = \sqrt{s} B_1^n$. The result follows from Theorem 10.2.1 and the bound (7.18) on the Gaussian width of the ℓ_1 ball:

$$w(T) = \sqrt{s}\, w(B_1^n) \le C \sqrt{s \log n}. \quad \square$$

Remark 10.3.5. The recovery error guaranteed by Corollary 10.3.4 is small if

$$m \gtrsim s \log n$$

(if the hidden constant here is appropriately large). In words, recovery is possible if *the number of measurements m is almost linear in the sparsity s*, while its dependence on the ambient dimension n is mild (logarithmic). This is good news. It means that for sparse signals, one can solve recovery problems in the high-dimensional regime where

$$m \ll n,$$

i.e. with much fewer measurements than the dimension.

Exercise 10.3.6 (Sparse recovery by convex optimization)**.** ☕☕☕

(a) Show that an unknown s-sparse signal x (without restriction on the norm) can be approximately recovered by solving the convex optimization problem

$$\text{minimize } \|x'\|_1 \text{ s.t. } y = Ax'. \tag{10.7}$$

The recovery error satisfies

$$\mathbb{E}\,\|\widehat{x} - x\|_2 \le CK^2 \sqrt{\frac{s \log n}{m}}\,\|x\|_2.$$

(b) Argue that a similar result holds for approximately sparse signals. State and prove such a guarantee.

10.3.3 The convex hull of sparse vectors, and the logarithmic improvement

The replacement of s-sparse vectors by the octahedron $\sqrt{s}B_1^n$ that we made in Exercise 10.3.3 is almost sharp. In the following exercise, we show that the convex hull of the set of sparse vectors

$$S_{n,s} := \left\{x \in \mathbb{R}^n : \ \|x\|_0 \le s,\ \|x\|_2 \le 1\right\}$$

is approximately the truncated ℓ_1 ball

$$T_{n,s} := \sqrt{s}B_1^n \cap B_2^n = \left\{x \in \mathbb{R}^n : \ \|x\|_1 \le \sqrt{s},\ \|x\|_2 \le 1\right\}.$$

Exercise 10.3.7 (The convex hull of sparse vectors)**.**

(a) Check that

$$\operatorname{conv}(S_{n,s}) \subset T_{n,s}.$$

(b) To help us prove a reverse inclusion, fix $x \in T_{n,s}$ and partition the support of x into disjoint subsets $I_1, I_2, \ldots$ so that I_1 indexes the s largest coefficients of x in magnitude, I_2 indexes the next s largest coefficients, and so on. Show that

$$\sum_{i \ge 1} \|x_{I_i}\|_2 \le 2,$$

where $x_I \in \mathbb{R}^T$ denotes the restriction of x onto a set I.

Hint: Note that $\|x_{I_1}\|_2 \le 1$. Next, for $i \ge 2$, note that each coordinate of x_{I_i} is smaller in magnitude than the average coordinate of $x_{I_{i-1}}$; conclude that $\|x_{I_i}\|_2 \le (1/\sqrt{s})\|x_{I_{i-1}}\|_1$. Then sum up the bounds.

(c) Deduce from part 2 that

$$T_{n,s} \subset 2\operatorname{conv}(S_{n,s}).$$

Exercise 10.3.8 (Gaussian width of the set of sparse vectors)**.** Use Exercise 10.3.7 to show that

$$w(T_{n,s}) \le 2w(S_{n,s}) \le C\sqrt{s \log(en/s)}.$$

Improve the logarithmic factor in the error bound for sparse recovery (Corollary 10.3.4) to

$$\mathbb{E}\,\|\widehat{x} - x\|_2 \le C\sqrt{\frac{s\log(en/s)}{m}}.$$

This shows that

$$m \gtrsim s\log(en/s)$$

measurements suffice for sparse recovery.

Exercise 10.3.9 (Sharpness). ☕☕☕☕ Show that

$$w(T_{n,s}) \ge w(S_{n,s}) \ge c\sqrt{s\log(2n/s)}.$$

Hint: Construct a large ε-separated subset of $S_{n,s}$ and thus deduce a lower bound on the covering numbers of $S_{n,s}$. Then use Sudakov's minoration inequality (Theorem 7.4.1 .

Exercise 10.3.10 (Garnaev-Gluskin's theorem). ☕☕☕ Improve the logarithmic factor in the bound (9.4.5) on the sections of the ℓ_1 ball. Namely, show that

$$\mathbb{E}\,\mathrm{diam}(B_1^n \cap E) \lesssim \sqrt{\frac{\log(en/m)}{m}}.$$

In particular, this shows that the logarithmic factor in (9.16) is not needed.
Hint: Fix $\rho > 0$ and apply the M^* bound for the truncated octahedron $T_\rho := B_1^n \cap \rho B_2^n$. Use Exercise 10.3.8 to bound the Gaussian width of T_ρ. Furthermore, note that if $\mathrm{rad}(T_\rho \cap E) \le \delta$ for some $\delta \le \rho$ then $\mathrm{rad}(T \cap E) \le \delta$. Finally, optimize in ρ.

10.4 Low-rank matrix recovery

In the following series of exercises, we establish a *matrix* version of the sparse recovery problem studied in Section 10.3. The unknown signal will now be a $d \times d$ matrix X instead of a signal $x \in \mathbb{R}^n$ considered previously.

There are two natural notions of sparsity for matrices. One is where most of the entries of X are zero, and this can be quantified by the ℓ_0 "norm" $\|X\|_0$, which counts non-zero entries. For this notion, we can directly apply the analysis of sparse recovery from Section 10.3. Indeed, it is enough to vectorize the matrix X and think of it as a long vector in $\mathbb{R}^{d^2}$.

But in this section, we consider an alternative and equally useful notion of sparsity for matrices: *low rank*. It is quantified by the rank of X, which we may think of as the ℓ_0 norm of the vector of the singular values of X, i.e.

$$s(X) := (s_i(X))_{i=1}^d. \tag{10.8}$$

Our analysis of the low-rank matrix recovery problem will roughly go along the same lines as the analysis of sparse recovery, but will not be identical to it.

Let us set up a low-rank matrix recovery problem. We would like to recover an unknown $d \times d$ matrix from m random measurements of the form

$$y_i = \langle A_i, X\rangle, \quad i = 1, \ldots, m. \tag{10.9}$$

Here A_i are independent $d \times d$ matrices, and $\langle A_i, X \rangle = \operatorname{tr}(A_i^\mathsf{T} X)$ is the canonical inner product of matrices (recall Section 4.1.3). In dimension $d = 1$, the matrix recovery problem (10.9) reduces to the vector recovery problem (10.2).

Since we have m linear equations in $d \times d$ variables, the matrix recovery problem is *ill-posed* if

$$m < d^2.$$

To be able to solve it in this range, we make an additional assumption that X has low rank, i.e.

$$\operatorname{rank}(X) \le r \ll d.$$

10.4.1 The nuclear norm

Like sparsity, the rank is not a convex function. To fix this, in Section 10.3 we replaced the sparsity (i.e. the ℓ_0 "norm") by the ℓ_1 norm. Let us try to do the same for the notion of rank. The rank of X is the ℓ_0 "norm" of the vector $s(X)$ of the singular values in (10.8). Replacing the ℓ_0 norm by the ℓ_1 norm, we obtain the quantity

$$\|X\|_* := \|s(X)\|_1 = \sum_{i=1}^d s_i(X) = \operatorname{tr}\left(\sqrt{X^\mathsf{T} X}\right)$$

which is called the *nuclear norm*, or *trace norm*, of X. (We omit the absolute values since the singular values are non-negative.)

Exercise 10.4.1. ☕☕☕ Prove that $\|\cdot\|_*$ is indeed a norm on the space of $d \times d$ matrices.

Hint: This will follow once you check the identity $\|X\|_* = \max\left\{ |\langle X, U \rangle| : U \in O(d) \right\}$ where $O(d)$ denotes the set of $d \times d$ orthogonal matrices. Prove the identity using the singular value decomposition of X.

Exercise 10.4.2 (Nuclear, Frobenius and operator norms)**.** ☕☕ Check that

$$\langle X, Y \rangle \le \|X\|_* \cdot \|Y\| \tag{10.10}$$

Conclude that

$$\|X\|_F^2 \le \|X\|_* \cdot \|X\|.$$

Hint: Think of the nuclear norm $\|\cdot\|_*$, Frobenius norm $\|\cdot\|_F$ and the operator norm $\|\cdot\|$ as matrix analogs of the ℓ_1 norm, ℓ_2 norm and ℓ_∞ norms for vectors, respectively.

Denote the unit ball of the nuclear norm by

$$B_* := \left\{ X \in \mathbb{R}^{d \times d} : \|X\|_* \le 1 \right\}.$$

Exercise 10.4.3 (Gaussian width of the unit ball of the nuclear norm)**.** ☕ Show that

$$w(B_*) \le 2\sqrt{d}.$$

Hint: Use (10.10) followed by Theorem 7.3.1.

The following is a matrix version of Exercise 10.3.3.

Exercise 10.4.4. Check that

$$\Big\{X \in \mathbb{R}^{d\times d} : \ \mathrm{rank}(X) \le r, \ \|X\|_F \le 1\Big\} \subset \sqrt{r} B_*.$$

10.4.2 Guarantees for low-rank matrix recovery

It makes sense to try to solve the low-rank matrix recovery problem (10.9) using the matrix version of the convex program (10.6), i.e.

$$\text{Find } X' : \ y_i = \langle A_i, X' \rangle \ \forall i = 1, \ldots, m; \quad \|X'\|_* \le \sqrt{r}. \tag{10.11}$$

Exercise 10.4.5 (Low-rank matrix recovery: guarantees)**.** Suppose the random matrices A_i are independent and have all independent, sub-gaussian entries.[1] Assume the unknown $d \times d$ matrix X with rank r satisfies $\|X\|_F \le 1$. Show that X can be approximately recovered from the random measurements y_i by a solution $\widehat{X}$ of the program (10.11). Show that the recovery error satisfies

$$\mathbb{E}\,\|\widehat{X} - X\|_F \le CK^2 \sqrt{\frac{rd}{m}}.$$

Remark 10.4.6. The recovery error becomes small if

$$m \gtrsim rd,$$

if the hidden constant here is appropriately large. This allows us to recover low-rank matrices even when the number of measurements m is too small, i.e. when

$$m \ll d^2$$

and the matrix recovery problem (without rank assumption) is ill-posed.

Exercise 10.4.7. Extend the matrix recovery result for *approximately* low-rank matrices.

The following is a matrix version of Exercise 10.3.6.

Exercise 10.4.8 (Low-rank matrix recovery by convex optimization)**.** Show that an unknown matrix X of rank r can be approximately recovered by solving the convex optimization problem

$$\text{minimize } \|X'\|_* \text{ s.t. } y_i = \langle A_i, X' \rangle \ \forall i = 1, \ldots, m.$$

Exercise 10.4.9 (Rectangular matrices)**.** Extend the matrix recovery result from quadratic to rectangular, $d_1 \times d_2$ matrices.

[1] The independence of entries can be relaxed. How?

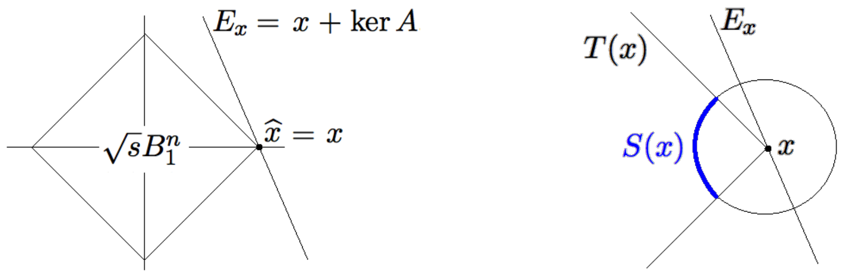
10.5 Exact recovery and the restricted isometry property

It turns out that the guarantees for sparse recovery we just developed can be dramatically improved: the recovery error for sparse signals x can actually be *zero*! We discuss two approaches to this remarkable phenomenon. First we deduce exact recovery from Escape Theorem 9.4.7. Next we present a general deterministic condition on a matrix A which guarantees exact recovery; it is known as the restricted isometry property (RIP). We check that random matrices A satisfy RIP, which gives another approach to exact recovery.

10.5.1 Exact recovery based on the Escape Theorem

To see why exact recovery should be possible, let us look at the recovery problem from a geometric viewpoint illustrated by Figure 10.3. A solution $\widehat{x}$ of the program (10.6) must lie in the intersection of the prior set T, which in our case is the ℓ_1 ball $\sqrt{s}B_1^n$, and the affine subspace $E_x = x + \ker A$.

The ℓ_1 ball is a polytope, and the s-sparse unit vector x lies on the $s-1$-dimensional edge of that polytope, see Figure 10.5a.



(a) Exact sparse recovery happens when the random subspace E_x is tangent to the ℓ_1 ball at the point x.

(b) The tangency occurs iff E_x is disjoint from the spherical part $S(x)$ of the tangent cone $T(x)$ of the ℓ_1 ball at point x.

Figure 10.5 Exact sparse recovery

It could happen with non-zero probability that the random subspace E_x is *tangent* to the polytope at the point x. If this does happen, x is the only point of intersection between the ℓ_1 ball and E_x. In this case, it follows that the solution $\widehat{x}$ to the program (10.6) is exact:

$$\widehat{x} = x.$$

To justify this argument, all we need to check is that a random subspace E_x is tangent to the ℓ_1 ball with high probability. We can do this using Escape Theorem 9.4.7. To see a connection, look at what happens in a small neighborhood around the tangent point, see Figure 10.5b. The subspace E_x is tangent if and only if the *tangent cone* $T(x)$ (formed by all rays emanating from x toward the points in the ℓ_1 ball) intersects E_x at a single point x. Equivalently, this happens if and only if the *spherical part* $S(x)$ of the cone (the intersection of $T(x)$ with a small sphere centered at x) is disjoint from E_x. But this is exactly the conclusion of Escape Theorem 9.4.7!

Let us now formally state the exact recovery result. We shall consider the noiseless sparse recovery problem

$$y = Ax.$$

and try to solve it using the optimization program (10.7), i.e.

$$\text{minimize } \|x'\|_1 \text{ s.t. } y = Ax'. \tag{10.12}$$

Theorem 10.5.1 (Exact sparse recovery)**.** *Suppose the rows A_i of A are independent, isotropic and sub-gaussian random vectors, and let $K := \max_i \|A_i\|_{\psi_2}$. Then the following happens with probability at least $1 - 2\exp(-cm/K^4)$.*

Assume an unknown signal $x \in \mathbb{R}^n$ is s-sparse and the number of measurements m satisfies

$$m \ge CK^4 s \log n.$$

Then a solution $\widehat{x}$ of the program (10.12) is exact, i.e.

$$\widehat{x} = x.$$

To prove the theorem, we would like to show that the recovery error

$$h := \widehat{x} - x$$

is zero. Let us examine the vector h more closely. First we show that h has more "energy" on the support of x than outside it.

Lemma 10.5.2. *Let $S := \operatorname{supp}(x)$. Then*

$$\|h_{S^c}\|_1 \le \|h_S\|_1.$$

Here $h_S \in \mathbb{R}^S$ denotes the restriction of the vector $h \in \mathbb{R}^n$ onto a subset of coordinates $S \subset \{1, \dots, n\}$.

Proof Since $\widehat{x}$ is the minimizer in the program (10.12), we have

$$\|\widehat{x}\|_1 \le \|x\|_1. \tag{10.13}$$

But there is also a lower bound

$$\begin{aligned} \|\widehat{x}\|_1 = \|x + h\|_1 &= \|x_S + h_S\|_1 + \|x_{S^c} + h_{S^c}\|_1 \\ &\ge \|x\|_1 - \|h_S\|_1 + \|h_{S^c}\|_1, \end{aligned}$$

where the last line follows by triangle inequality and using $x_S = x$ and $x_{S^c} = 0$. Substitute this bound into (10.13) and cancel $\|x\|_1$ on both sides to complete the proof. □

Lemma 10.5.3. *The error vector satisfies*

$$\|h\|_1 \le 2\sqrt{s}\|h\|_2.$$

Proof Using Lemma 10.5.2 and then Cauchy-Schwarz inequality, we obtain

$$\|h\|_1 = \|h_S\|_1 + \|h_{S^c}\|_1 \le 2\|h_S\|_1 \le 2\sqrt{s}\|h_S\|_2.$$

Since trivially $\|h_S\|_2 \le \|h\|_2$, the proof is complete. □

Proof of Theorem 10.5.1 Assume that the recovery is not exact, i.e.

$$h = \widehat{x} - x \ne 0.$$

By Lemma 10.5.3, the normalized error $h/\|h\|_2$ lies in the set

$$T_s := \left\{ z \in S^{n-1} : \ \|z\|_1 \le 2\sqrt{s} \right\}.$$

Since also

$$Ah = A\widehat{x} - Ax = y - y = 0,$$

we have

$$\frac{h}{\|h\|_2} \in T_s \cap \ker A. \tag{10.14}$$

Escape Theorem 9.4.7 states that this intersection is empty with high probability, as long as

$$m \ge CK^4 w(T_s)^2.$$

Now,

$$w(T_s) \le 2\sqrt{s}w(B_1^n) \le C\sqrt{s\log n}, \tag{10.15}$$

where we used the bound (7.18) on the Gaussian width of the ℓ_1 ball. Thus, if $m \ge CK^4 s \log n$, the intersection in (10.14) is empty with high probability, which means that the inclusion in (10.14) can not hold. This contradiction implies that our assumption that $h \ne 0$ is false with high probability. The proof is complete. □

Exercise 10.5.4 (Improving the logarithmic factor)**.** ☕ Show that the conclusion of Theorem 10.5.1 holds under a weaker assumption on the number of measurements, which is

$$m \ge CK^4 s \log(en/s).$$

Hint: Use the result of Exercise 10.3.8.

Exercise 10.5.5. ☕☕ Give a geometric interpretation of the proof of Theorem 10.5.1, using Figure 10.5b. What does the proof say about the tangent cone $T(x)$? Its spherical part $S(x)$?

Exercise 10.5.6 (Noisy measurements)**.** ☕☕☕ Extend the result on sparse recovery (Theorem 10.5.1) for noisy measurements, where

$$y = Ax + w.$$

You may need to modify the recovery program by making the constraint $y = Ax'$ approximate?

Remark 10.5.7. Theorem 10.5.1 shows that one can effectively solve *under-determined systems of linear equations* $y = Ax$ with $m \ll n$ equations in n variables, if the solution is sparse.

10.5.2 Restricted isometries

This subsection is optional; the further material is not based on it.

All recovery results we proved so far were probabilistic: they were valid for a random measurement matrix A and with high probability. We may wonder if there exists a *deterministic* condition which can guarantee that a given matrix A can be used for sparse recovery. Such condition is the restricted isometry property (RIP).

Definition 10.5.8 (RIP)**.** An $m \times n$ matrix A satisfies the *restricted isometry property* (RIP) with parameters α, β and s if the inequality

$$\alpha \|v\|_2 \le \|Av\|_2 \le \beta \|v\|_2$$

holds for all vectors $v \in \mathbb{R}^n$ such that[2] $\|v\|_0 \le s$.

In other words, a matrix A satisfies RIP if the restriction of A on any s-dimensional coordinate subspace of $\mathbb{R}^n$ is an approximate isometry in the sense of (4.5).

Exercise 10.5.9 (RIP via singular values)**.** ☕ Check that RIP holds if and only if the singular values satisfy the inequality

$$\alpha \le s_s(A_I) \le s_1(A_I) \le \beta$$

for all subsets $I \subset [n]$ of size $|I| = s$. Here A_I denotes the $m \times s$ sub-matrix of A formed by selecting the columns indexed by I.

Now we prove that RIP is indeed a sufficient condition for sparse recovery.

Theorem 10.5.10 (RIP implies exact recovery)**.** *Suppose an $m \times n$ matrix A satisfies RIP with some parameters α, β and $(1 + \lambda)s$, where $\lambda > (\beta/\alpha)^2$. Then every s-sparse vector $x \in \mathbb{R}^n$ can be recovered exactly by solving the program (10.12), i.e. the solution satisfies*

$$\widehat{x} = x.$$

Proof As in the proof of Theorem 10.5.1, we would like to show that the recovery error

$$h = \widehat{x} - x$$

is zero. To do this, we decompose h in a way similar to Exercise 10.3.7.

Step 1: decomposing the support. Let I_0 be the support of x; let I_1 index the λs largest coefficients of $h_{I_0^c}$ in magnitude; let I_2 index the next λs largest coefficients of $h_{I_0^c}$ in magnitude, and so on. Finally, denote $I_{0,1} = I_0 \cup I_1$.

[2] Recall from Section 10.3.1 that by $\|v\|_0$ we denote the number of non-zero coordinates of v.

Since

$$Ah = A\widehat{x} - Ax = y - y = 0,$$

triangle inequality yields

$$0 = \|Ah\|_2 \geq \|A_{I_{0,1}} h_{I_{0,1}}\|_2 - \|A_{I_{0,1}^c} h_{I_{0,1}^c}\|_2. \tag{10.16}$$

Next, we examine the two terms in the right side.

Step 2: applying RIP. Since $|I_{0,1}| \leq s + \lambda s$, RIP yields

$$\|A_{I_{0,1}} h_{I_{0,1}}\|_2 \geq \alpha \|h_{I_{0,1}}\|_2$$

and triangle inequality followed by RIP also give

$$\|A_{I_{0,1}^c} h_{I_{0,1}^c}\|_2 \leq \sum_{i \geq 2} \|A_{I_i} h_{I_i}\|_2 \leq \beta \sum_{i \geq 2} \|h_{I_i}\|_2.$$

Plugging into (10.16) gives

$$\beta \sum_{i \geq 2} \|h_{I_i}\|_2 \geq \alpha \|h_{I_{0,1}}\|_2. \tag{10.17}$$

Step 3: summing up. Next, we bound the sum in the left like we did in Exercise 10.3.7. By definition of I_i, each coefficient of h_{I_i} is bounded in magnitude by the average of the coefficients of $h_{I_{i-1}}$, i.e. by $\|h_{I_{i-1}}\|_1/(\lambda s)$ for $i \geq 2$. Thus

$$\|h_{I_i}\|_2 \leq \frac{1}{\sqrt{\lambda s}} \|h_{I_{i-1}}\|_1.$$

Summing up, we get

$$\begin{aligned}
\sum_{i \geq 2} \|h_{I_i}\|_2 &\leq \frac{1}{\sqrt{\lambda s}} \sum_{i \geq 1} \|h_{I_i}\|_1 = \frac{1}{\sqrt{\lambda s}} \|h_{I_0^c}\|_1 \\
&\leq \frac{1}{\sqrt{\lambda s}} \|h_{I_0}\|_1 \quad \text{(by Lemma 10.5.2)} \\
&\leq \frac{1}{\sqrt{\lambda}} \|h_{I_0}\|_2 \leq \frac{1}{\sqrt{\lambda}} \|h_{I_{0,1}}\|_2.
\end{aligned}$$

Putting this into (10.17) we conclude that

$$\frac{\beta}{\sqrt{\lambda}} \|h_{I_{0,1}}\|_2 \geq \alpha \|h_{I_{0,1}}\|_2.$$

This implies that $h_{I_{0,1}} = 0$ since $\beta/\sqrt{\lambda} < \alpha$ by assumption. By construction, $I_{0,1}$ contains the largest coefficient of h. It follows that $h = 0$ as claimed. The proof is complete. □

Unfortunately, it is unknown how to construct deterministic matrices A that satisfy RIP with good parameters (i.e. with $\beta = O(\alpha)$ and with s as large as m, up to logarithmic factors). However, it is quite easy to show that random matrices A do satisfy RIP with high probability:

Theorem 10.5.11 (Random matrices satisfy RIP)**.** *Consider an $m \times n$ matrix A whose rows A_i of A are independent, isotropic and sub-gaussian random vectors, and let $K := \max_i \|A_i\|_{\psi_2}$. Assume that*

$$m \geq CK^4 s \log(en/s).$$

Then, with probability at least $1 - 2\exp(-cm/K^4)$, the random matrix A satisfies RIP with parameters $\alpha = 0.9\sqrt{m}$, $\beta = 1.1\sqrt{m}$ and s.

Proof By Exercise 10.5.9, it is enough to control the singular values of all $m \times s$ sub-matrices A_I. We will do it by using the two-sided bound from Theorem 4.6.1 and then taking the union bound over all sub-matrices.

Let us fix I. Theorem 4.6.1 yields

$$\sqrt{m} - r \leq s_s(A_I) \leq s_1(A_I) \leq \sqrt{m} + r$$

with probability at least $1 - 2\exp(-t^2)$, where $r = C_0K^2(\sqrt{s} + t)$. If we set $t = \sqrt{m}/(20C_0K^2)$ and use the assumption on m with appropriately large constant C, we can make sure that $r \leq 0.1\sqrt{m}$. This yields

$$0.9\sqrt{m} \leq s_s(A_I) \leq s_1(A_I) \leq 1.1\sqrt{m} \tag{10.18}$$

with probability at least $1 - 2\exp(-2cm^2/K^4)$, where $c > 0$ is an absolute constant.

It remains to take a union bound over all s-element subsets $I \subset [n]$; there are $\binom{n}{s}$ of them. We conclude that (10.18) holds with probability at least

$$1 - 2\exp(-2cm^2/K^4) \cdot \binom{n}{s} > 1 - 2\exp(-cm^2/K^4).$$

To get the last inequality, recall that $\binom{n}{s} \leq \exp(s\log(en/s))$ by (0.0.5) and use the assumption on m. The proof is complete. □

The results we just proved give another approach to Theorem 10.5.1 about exact recovery with a random matrix A.

Second proof of Theorem 10.5.1 By Theorem 10.5.11, A satisfies RIP with $\alpha = 0.9\sqrt{m}$, $\beta = 1.1\sqrt{m}$ and $3s$. Thus, Theorem 10.5.10 for $\lambda = 2$ guarantees exact recovery. We conclude that Theorem 10.5.1 holds, and we even get the logarithmic improvement noted in Exercise 10.5.4. □

An advantage of RIP is that this property is often simpler to verify than to prove exact recovery directly. Let us give one example.

Exercise 10.5.12 (RIP for random projections)**.** ☕☕☕ Let P be the orthogonal projection in $\mathbb{R}^n$ onto an m-dimensional random subspace uniformly distributed in the Grassmanian $G_{n,m}$.

(a) Prove that P satisfies RIP with good parameters (similar to Theorem 10.5.11, up to a normalization).
(b) Conclude a version of Theorem 10.5.1 for exact recovery from random projections.

10.6 Lasso algorithm for sparse regression

In this section we analyze an alternative method for sparse recovery. This method was originally developed in statistics for the equivalent problem of *sparse linear regression*, and it is called Lasso ("least absolute shrinkage and selection operator").

10.6.1 Statistical formulation

Let us recall the classical linear regression problem, which we described in Example 10.1.2. It is

$$Y = X\beta + w \tag{10.19}$$

where X is a known $m \times n$ matrix that contains a sample of predictor variables, $Y \in \mathbb{R}^m$ is a known vector that contains a sample of the values of the response variable, $\beta \in \mathbb{R}^n$ is an unknown coefficient vector that specifies the relationship between predictor and response variables, and w is a noise vector. We would like to recover β.

If we do not assume anything else, the regression problem can be solved by the method of *ordinary least squares*, which minimizes the ℓ_2-norm of the error over all candidates for β:

$$\text{minimize } \|Y - X\beta'\|_2 \text{ s.t. } \beta' \in \mathbb{R}^n. \tag{10.20}$$

Now let us make an extra assumption that β' is *sparse*, so that the response variable depends only on a few of the n predictor variables (e.g. the cancer depends on few genes). So, like in (10.5), we assume that

$$\|\beta\|_0 \le s$$

for some $s \ll n$. As we argued in Section 10.3, the ℓ_0 is not convex, and its convex proxy is the ℓ_1 norm. This prompts us to modify the ordinary least squares program (10.20) by including a restriction on the ℓ_1 norm, which promotes sparsity in the solution:

$$\text{minimize } \|Y - X\beta'\|_2 \text{ s.t. } \|\beta'\|_1 \le R, \tag{10.21}$$

where R is a parameter which specifies a desired sparsity level of the solution. The program (10.21) is one of the formulations of Lasso, the most popular statistical method for sparse linear regression. It is a convex program, and therefore is computationally tractable.

10.6.2 Mathematical formulation and guarantees

It would be convenient to return to the notation we used for sparse recovery instead of using the statistical notation in the previous section. So let us restate the linear regression problem (10.19) as

$$y = Ax + w$$

where A is a known $m \times n$ matrix, $y \in \mathbb{R}^m$ is a known vector, $x \in \mathbb{R}^n$ is an unknown vector that we are trying to recover, and $w \in \mathbb{R}^m$ is noise which is either fixed or random and independent of A. Then Lasso program (10.21) becomes

$$\text{minimize } \|y - Ax'\|_2 \text{ s.t. } \|x'\|_1 \le R. \tag{10.22}$$

We prove the following guarantee of the performance of Lasso.

Theorem 10.6.1 (Performance of Lasso)**.** *Suppose the rows A_i of A are independent, isotropic and sub-gaussian random vectors, and let $K := \max_i \|A_i\|_{\psi_2}$. Then the following happens with probability at least $1 - 2\exp(-s \log n)$.*

Assume an unknown signal $x \in \mathbb{R}^n$ is s-sparse and the number of measurements m satisfies

$$m \ge CK^4 s \log n. \tag{10.23}$$

Then a solution $\widehat{x}$ of the program (10.22) with $R := \|x\|_1$ is accurate, namely

$$\|\widehat{x} - x\|_2 \le C\sigma\sqrt{\frac{s \log n}{m}},$$

where $\sigma = \|w\|_2/\sqrt{m}$.

Remark 10.6.2 (Noise)**.** The quantity σ^2 is the *average squared noise per measurement*, since

$$\sigma^2 = \frac{\|w\|_2^2}{m} = \frac{1}{m}\sum_{i=1}^{m} w_i^2.$$

Then, if the number of measurements is

$$m \gtrsim s \log n,$$

Theorem 10.6.1 bounds the recovery error by the average noise per measurement σ. And if m is larger, the recovery error gets smaller.

Remark 10.6.3 (Exact recovery)**.** In the noiseless model $y = Ax$ we have $w = 0$ and thus Lasso recovers x exactly, i.e.

$$\widehat{x} = x.$$

The proof of Theorem 10.6.1 will be similar to our proof of Theorem 10.5.1 on exact recovery, although instead of the Escape theorem we use Matrix Deviation Inequality (Theorem 9.1.1) directly this time.

We would like to bound the norm of the error vector

$$h := \widehat{x} - x.$$

Exercise 10.6.4. ☕☕ Check that h satisfies the conclusions of Lemmas 10.5.2 and 10.5.3, so we have

$$\|h\|_1 \le 2\sqrt{s}\|h\|_2. \tag{10.24}$$

Hint: The proofs of these lemmas are based on the fact that $\|\widehat{x}\|_1 \le \|x\|_1$, which holds in our situation as well.

In case where the noise w is nonzero, we can not expect to have $Ah = 0$ like in Theorem 10.5.1. (Why?) Instead, we can give an upper and a lower bounds for $\|Ah\|_2$.

Lemma 10.6.5 (Upper bound on $\|Ah\|_2$). *We have*

$$\|Ah\|_2^2 \le 2 \left\langle h, A^\mathsf{T} w \right\rangle. \tag{10.25}$$

Proof Since $\widehat{x}$ is the minimizer of Lasso program (10.22), we have

$$\|y - A\widehat{x}\|_2 \le \|y - Ax\|_2.$$

Let us express both sides of this inequality in terms of h and w, using that $y = Ax + w$ and $h = \widehat{x} - x$:

$$\begin{aligned} &y - A\widehat{x} = Ax + w - A\widehat{x} = w - Ah; \\ &y - Ax = w. \end{aligned}$$

So we have

$$\|w - Ah\|_2 \le \|w\|_2.$$

Square both sides:

$$\|w\|_2^2 - 2\langle w, Ah\rangle + \|Ah\|_2^2 \le \|w\|_2^2.$$

Simplifying this bound completes the proof. □

Lemma 10.6.6 (Lower bound on $\|Ah\|_2$). *With probability at least* $1-2\exp(-4s\log n)$, *we have*

$$\|Ah\|_2^2 \ge \frac{m}{4}\|h\|_2^2.$$

Proof By (10.24), the normalized error $h/\|h\|_2$ lies in the set

$$T_s := \left\{ z \in S^{n-1} : \ \|z\|_1 \le 2\sqrt{s} \right\}.$$

Use matrix deviation inequality in its high-probability form (Exercise 9.1.8) with $u = 2\sqrt{s\log n}$. It yields that, with probability at least $1 - 2\exp(-4s\log n)$,

$$\begin{aligned} \sup_{z \in T_s} \left| \|Az\|_2 - \sqrt{m} \right| &\le C_1 K^2 \left(w(T_s) + 2\sqrt{s\log n} \right) \\ &\le C_2 K^2 \sqrt{s\log n} \quad \text{(recalling (10.15))} \\ &\le \frac{\sqrt{m}}{2} \quad \text{(by assumption on } m\text{)}. \end{aligned}$$

To make the last line work, choose the absolute constant C in (10.23) large enough. By triangle inequality, this implies that

$$\|Az\|_2 \ge \frac{\sqrt{m}}{2} \quad \text{for all } z \in T_s.$$

Substituting $z := h/\|h\|_2$, we complete the proof. □

The last piece we need to prove Theorem 10.6.1 is an upper bound on the right hand side of (10.25).

Lemma 10.6.7. *With probability at least* $1 - 2\exp(-4s\log n)$*, we have*

$$\left\langle h, A^\mathsf{T} w \right\rangle \le CK\|h\|_2\|w\|_2\sqrt{s\log n}. \tag{10.26}$$

Proof As in the proof of Lemma 10.6.6, the normalized error satisfies

$$z = \frac{h}{\|h\|_2} \in T_s.$$

So, dividing both sides of (10.26) by $\|h\|_2$, we see that it is enough to bound the supremum random process

$$\sup_{z\in T_s} \left\langle z, A^\mathsf{T} w \right\rangle$$

with high probability. We are going to use Talagrand's comparison inequality (Corollary 8.6.3). This result applies for random processes with sub-gaussian increments, so let us check this condition first.

Exercise 10.6.8. ☕☕ Show that the random process

$$X_t := \left\langle t, A^\mathsf{T} w \right\rangle, \quad t \in \mathbb{R}^n,$$

has sub-gaussian increments, and

$$\|X_t - X_s\|_{\psi_2} \le CK\|w\|_2 \cdot \|t-s\|_2.$$

Hint: Recall the proof of sub-gaussian Chevet's inequality (Theorem 8.7.1).

Now we can use Talagrand's comparison inequality in the high-probability form (Exercise 8.6.5) for $u = 2\sqrt{s\log n}$. We obtain that, with probability at least $1 - 2\exp(-4s\log n)$,

$$\begin{aligned}\sup_{z\in T_s} \left\langle z, A^\mathsf{T} w \right\rangle &\le C_1 K\|w\|_2 \left(w(T_s) + 2\sqrt{s\log n}\right) \\ &\le C_2 K\|w\|_2\sqrt{s\log n} \quad \text{(recalling (10.15))}.\end{aligned}$$

This completes the proof of Lemma 10.6.7. □

Proof of Theorem 10.6.1. Put together the bounds in Lemmas 10.6.5, 10.6.6 and 10.26. By union bound, we have that with probability at least $1-4\exp(-4s\log n)$,

$$\frac{m}{4}\|h\|_2^2 \le CK\|h\|_2\|w\|_2\sqrt{s\log n}.$$

Solving for $\|h\|_2$, we obtain

$$\|h\|_2 \le CK\frac{\|w\|_2}{\sqrt{m}} \cdot \sqrt{\frac{s\log n}{m}}.$$

This completes the proof of Theorem 10.6.1. □

Exercise 10.6.9 (Improving the logarithmic factor). ☕ Show that Theorem 10.6.1 holds if $\log n$ is replaced by $\log(en/s)$, thus giving a stronger guarantee.
Hint: Use the result of Exercise 10.3.8.

Exercise 10.6.10. ☕☕ Deduce the exact recovery guarantee (Theorem 10.5.1) directly from the Lasso guarantee (Theorem 10.6.1). The probability that you get could be a bit weaker.

Another popular form of Lasso program (10.22) is the following *unconstrained version*:

$$\text{minimize } \|y - Ax'\|_2^2 + \lambda\|x'\|_1, \tag{10.27}$$

This is a convex optimization problem, too. Here λ is a parameter which can be adjusted depending on the desired level of sparsity. The method of Lagrange multipliers shows that the constrained and unconstrained versions of Lasso are equivalent for appropriate R and λ. This however does not immediately tell us how to choose λ. The following exercise settles this question.

Exercise 10.6.11 (Unconstrained Lasso). ☕☕☕☕ Assume that the number of measurements satisfy

$$m \gtrsim s \log n.$$

Choose the parameter λ so that $\lambda \gtrsim \sqrt{\log n}\|w\|_2$. Then, with high probability, the solution $\widehat{x}$ of unconstrained Lasso (10.27) satisfies

$$\|\widehat{x} - x\|_2 \lesssim \frac{\lambda\sqrt{s}}{m}.$$

10.7 Notes

The applications we discussed in this chapter are drawn from two fields: signal processing (specifically, compressed sensing) and high-dimensional statistics (more precisely, high-dimensional structured regression). The tutorial [223] offers a unified treatment of these two kinds problems, which we followed in this chapter. The survey [56] and book [78] offer a deeper introduction into compressed sensing. The books [100, 42] discuss statistical aspects of sparse recovery.

Signal recovery based on M^* bound discussed in Section 10.2 is based on [223], which has various versions of Theorem 10.2.1 and Corollary 10.3.4. Garnaev-Gluskin's bound from Exercise 10.3.10 was first proved in [80], see also [136] and [78, Chapter 10].

The survey [58] offers a comprehensive overview of the low-rank matrix recovery problem, which we discussed in Section 10.4. Our presentation is based on [223, Section 10].

The phenomenon of exact sparse recovery we discussed in Section 10.5 goes back to the origins of compressed sensing; see [56] and book [78] for its history and recent developments. Our presentation of exact recovery via escape theorem in Section 10.5.1 partly follows [223, Section 9]; see also [52, 189] and especially

[208] for applications of the escape theorem to sparse recovery. One can obtain very precise guarantees that give asymptotically sharp formulas (the so-called phase transitions) for the number of measurements needed for signal recovery. The first such phase transitions were identified in [68] for sparse signals and uniform random projection matrices A; see also [67, 64, 65, 66]. More recent work clarified phase transitions for general feasible sets T and more general measurement matrices [9, 162, 163].

The approach to exact sparse recovery based on RIP presented in Section 10.5.2 was pioneered by E. Candes and T. Tao [46]; see [78, Chapter 6] for a comprehensive introduction. An early form of Theorem 10.5.10 already appear in [46]. The proof we gave here was communicated to the author by Y. Plan; it is similar to the argument of [44]. The fact that random matrices satisfy RIP (exemplified by Theorem 10.5.11) is a backbone of compressed sensing; see [78, Section 9.1, 12.5], [222, Section 5.6].

The Lasso algorithm for sparse regression that we studies in Section 10.6 was pioneered by R. Tibshirani [204]. The books [100, 42] offer a comprehensive introduction into statistical problems with sparsity constraints; these books discuss Lasso and its many variants. A version of Theorem 10.6.1 and some elements of its proof can be traced to the work of P. J. Bickel, Y. Ritov and A. Tsybakov [22], although their argument was not based on matrix deviation inequality. Theoretical analysis of Lasso is also presented in [100, Chapter 11] and [42, Chapter 6].